

991 + 981 , MY 2012 =>

Diagnosis information - Headlights left side

10000.....	1
10001.....	2
10003.....	3
20000.....	4
20000.....	5
20001.....	6
20001.....	7
20002.....	8
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20003.....	10
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30020.....	53
30021.....	54
30021.....	55
30300.....	56
30300.....	57
30301.....	58
320050.....	59
320050.....	60
320050.....	61
40000.....	62
40001.....	63
40002.....	64
40003.....	65
40004.....	66
40005.....	67
40006.....	68
40011.....	69
40020.....	70
40021.....	71
40300.....	72
40301.....	73
40400.....	74
50000.....	75
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Diagnostic information - Headlights

Control module Power supply

WARNING

HBA Xenon: Electric shock and burns from high voltage.

- ◆ The lights should always be switched off before removing and installing the headlights.
- ◆ Only operate the gas-discharge lamp when installed in the headlight.
- ◆ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Control module** Headlights **Power supply**
- Signal Terminal 15
- Signal Terminal 56b

Fault effects

- ◆ System switches off, Headlights **OFF**
- ◆ Display in instrument cluster

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- **Control module** Headlights **Power supply**
- Signal Terminal 15
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Fault effects

- ◆ System switches off, Headlights **OFF**
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Diagnostic information - Headlights

Coding plug Wrong connector assignment

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Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

Incorrect coding (pin assignment/contacts) of the headlight control unit connector could be the cause.

- **Control module** Headlights **Pin assignment** incorrect
- **Control module** Headlights **Pin assignment** interchanged

Fault effects

- ◆ System switches off
- ◆ Display in instrument cluster

HBA stepper motor



CAUTION

Danger of injury from high voltage!

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- ◆ The gas discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or the wiring when the part is removed and disconnected from the power supply.

Diagnostic conditions

- Headlight (xenon) control unit internal function test
- Ignition on

Possible fault causes

- ◆ Line connection or contacts to HBA servo motor faulty
- ◆ HBA stepper motor overheated
- ◆ HBA stepper motor mechanically blocked
- ◆ HBA stepper motor faulty
- ◆ **Headlight** control unit coded incorrectly

Stepper motor HBA switched off



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Possible fault causes

- ◆ HBA stepper motor version is not supported by control unit
- ◆ HBA stepper motor switched off due to overtemperature
- ◆ Supply voltage too low
- ◆ Headlight control unit wrong software version

HBA stepper motor



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- ◆ Headlight control unit wrong software version

Diagnostic information - Xenon headlights

Front compression angle sensor (only for Xenon without **leveling system control unit (PASM)**)

NOTE

- ♦ On vehicles with level control/PASM, the signal for the vehicle position is generated via the leveling system/PASM control unit. This has four angle sensors for compression.
- ♦ The signal values are then sent to the headlights via CAN
- ♦ Calibration takes place in the **leveling system control unit (PASM)**
- ♦ Calibration takes place in the left Xenon headlight only on vehicles without a **leveling system control unit (PASM)**

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Angle sensor** calibration incorrect or not carried out
- **>Commissioning<** function under Maintenance/repairs not performed
- **Angle sensor power supply**
- **Angle sensor** signal line faulty
- **Angle sensor** faulty

Fault effects

- ♦ System switches off
- ♦ Display in instrument cluster

Diagnostic information - Headlight control unit

Power supply to HBA angle sensor (not leveling system)

CAUTION

Danger of injury from high voltage!

- ♦ The light must always be switched off when removing and installing the headlights.
- ♦ The gas-discharge lamp may only be operated when installed in the headlight.
- ♦ Only carry out repairs to the headlight or wiring when the part is removed and disconnected from the power supply.

INFORMATION

- ♦ The angle sensor (compression) is also referred to as the vehicle level sensor.
- ♦ The power supply (5 V) for the angle sensors is supplied via the **left headlight control unit**.
- ♦ HBA (headlight beam adjustment) is also referred to as AHBA (automatic headlight beam adjustment)
- ♦ In the case of the leveling system, the signal comes from the leveling system control unit via CAN

Diagnostic conditions

- Internal function test on headlight (Xenon) control unit
- Ignition on

Possible fault causes

- ♦ Supply voltage line connection faulty, contacts faulty
- ♦ Ground supply line connection faulty, contacts faulty
- ♦ **Headlight** control unit coded incorrectly
- ♦ Line connection in headlight housing
- ♦ Headlight control unit (Xenon)

Diagnostic information - Headlight control unit

Power supply to HBA angle sensor (not leveling system)

CAUTION

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Diagnostic conditions

- Internal function test on headlight (Xenon) control unit
- Ignition on

Possible fault causes

- ♦ Supply voltage line connection faulty, contacts faulty
- ♦ Ground supply line connection faulty, contacts faulty
- ♦ **Headlight** control unit coded incorrectly
- ♦ Line connection in headlight housing
- ♦ Headlight control unit (Xenon)

Diagnostic information - Xenon headlights

Rear compression angle sensor (only for Xenon without **leveling system control unit (PASM)**)

INFORMATION

- ◆ On vehicles with level control/PASM, the signal for the vehicle position is generated via the leveling system/PASM control unit. This has four angle sensors for compression.
- ◆ The signal values are then sent to the headlights via CAN
- ◆ Calibration takes place in the **leveling system control unit (PASM)**
- ◆ Calibration takes place in the left Xenon headlight only on vehicles without a **leveling system control unit (PASM)**

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- **Angle sensor** calibration incorrect or not carried out
- **>Commissioning<** function under Maintenance/repairs not performed
- **Angle sensor power supply**
- **Angle sensor** signal line faulty
- **Angle sensor** faulty

Fault effects

- ◆ System switches off
- ◆ Display in instrument cluster

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Diagnostic information - Headlight control unit

Stepper motor for dyn. cornering light switched off

CAUTION

Danger of injury from high voltage!

- ◆ The light must always be switched off when removing and installing the headlights.
- ◆ The gas-discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or wiring when the part is removed and disconnected from the power supply.

INFORMATION

The stepper motor/servo motor for the dynamic cornering light and its mechanical components and wiring cannot be replaced individually.
The headlight housing must be replaced if repairs are needed.

Diagnostic conditions

- Internal function test on headlight (Xenon) control unit
- Ignition on

Possible fault causes

- ◆ Mechanism of stepper motor for dyn. cornering light sluggish or blocked
- ◆ Dynamic cornering light wiring faulty
- ◆ Stepper motor for dynamic cornering light faulty
- ◆ Headlight control unit faulty

Diagnostic information - Headlight control unit

Stepper motor for dyn. cornering light, position sensor

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- Ignition on

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- ◆ Mechanism of stepper motor for dyn. cornering light sluggish or blocked
- ◆ Dynamic cornering light wiring faulty
- ◆ Stepper motor for dynamic cornering light faulty
- ◆ Headlight control unit faulty

Diagnostic information - Headlight control unit

Stepper motor for dyn. cornering light switched off

CAUTION

Danger of injury from high voltage!

- ◆ The light must always be switched off when removing and installing the headlights.
- ◆ The gas-discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or wiring when the part is removed and disconnected from the power supply.

INFORMATION

The stepper motor/servo motor for the dynamic cornering light and its mechanical components and wiring cannot be replaced individually.
The headlight housing must be replaced if repairs are needed.

Diagnostic conditions

- Internal function test on headlight (Xenon) control unit
- Ignition on

Possible fault causes

- ◆ Mechanism of stepper motor for dyn. cornering light sluggish or blocked
- ◆ Dynamic cornering light wiring faulty
- ◆ Stepper motor for dynamic cornering light faulty
- ◆ Headlight control unit faulty

Diagnostic information - Headlight control unit

Stepper motor for dyn. cornering light, position sensor

CAUTION

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